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FFR Physiological Plausibility Scoring Using a Conditional GAN

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March 2026

Word Count: 6516

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1. Introduction

Coronary artery disease remains one of the leading causes of illness and death globally[1], and accurate assessment of stenosis severity is essential for guiding clinical intervention [1]. In this context, Fractional Flow Reserve (FFR) has a major role as a physiological indicator of whether a coronary narrowing is functionally significant [1]. As computational methods increasingly complement conventional diagnostic tools, ensuring the physiological validity of simulation-derived outputs has become an important challenge [2]. This project addresses that challenge by proposing a deep learning framework that learns the conditional pattern of physiologically consistent FFR values and automatically identifies samples that deviate from it.

1.1 Motivation and Aims

Fractional Flow Reserve (FFR) is widely regarded as a gold-standard physiological index for assessing the haemodynamic importance of coronary artery stenosis and supporting decisions regarding revascularisation [1]. Although invasive FFR measurement remains the clinical standard, Computational Fluid Dynamics (CFD) has emerged as a promising non-invasive alternative for estimating FFR from patient-specific anatomical models [2]. However, the reliability of CFD-derived FFR is highly sensitive to modelling assumptions and the way anatomical parameters are defined during simulation setup.

A particularly important issue arises when stenosis severity is defined inconsistently across datasets. In clinical practice, stenosis severity is commonly interpreted in terms of vessel diameter reduction, whereas some simulated datasets may instead define severity using cross-sectional area reduction. Although these definitions appear related, they are not equivalent and can lead to systematically different physiological outcomes. As a result, an FFR value that appears plausible under one definition may be physiologically inconsistent under another. This creates the risk of biased datasets containing anatomically plausible geometries but functionally implausible FFR values.

Because manual inspection of large datasets is impractical, there is a clear need for automated quality-control methods that can distinguish physiologically consistent samples from biased or out-of-distribution cases. The aim of this project is therefore to develop and evaluate a deep learning framework that learns the distribution of physiologically plausible FFR values from a trusted reference dataset and uses this learned representation to detect biased samples.
The primary objectives of the project are:

- to design a model capable of learning the complex conditional distribution of physiologically consistent FFR values from a trusted reference dataset;
- to use this learned distribution as the basis for an out-of-distribution detector that assigns a quantitative plausibility score to new samples;
- to evaluate whether the model can successfully identify physiologically implausible samples from a known biased dataset.

1.2 The Proposed Solution: A Conditional GAN for OOD Detection

To address these aims, this work proposes a conditional Generative Adversarial Network (cGAN) as a framework for physiological plausibility assessment. Rather than using the GAN primarily for synthetic data generation, the model is used to learn whether a given pairing of anatomical conditions and FFR value is consistent with the physiological distribution represented by the trusted reference data.

The model is conditioned on anatomical and lesion-related features such as artery diameter, stenosis severity, lesion length, and engineered haemodynamic descriptors. Training is performed only on the trusted reference dataset, while the biased dataset is reserved entirely for evaluation. Once trained, the discriminator acts as a standalone scoring function that assigns each FFR-anatomy pair a plausibility score between 0 and 1. Higher scores indicate stronger consistency with the learned physiological distribution, whereas lower scores suggest likely out-of-distribution behaviour associated with simulation bias.

1.3 Report Structure

This report is organised as follows. Section 2 reviews the relevant background literature on FFR, CFD, Generative Adversarial Networks, and their use in anomaly and out-of-distribution detection. Section 3 describes the full methodology, including dataset construction, preprocessing, feature engineering, model architecture, and the two-phase training strategy. Section 4 presents the experimental results, including overall discrimination performance, plausibility score behaviour, and threshold-based out-of-distribution detection. Section 5 discusses the interpretation of the results, justifies the principal design decisions, and outlines the limitations and future directions of the work. Finally, Section 6 concludes the report by summarising the key findings and contributions.

2. Background and Literature Review

This section provides an overview of the core concepts and technologies that form the foundation of this project. It begins by discussing the clinical and computational context of Fractional Flow Reserve, then introduces the deep learning architectures, Generative Adversarial Networks and their conditional variants, and concludes by reviewing their application to the problem of anomaly and out-of-distribution detection.

2.1 Fractional Flow Reserve (FFR) and CFD

In the clinical assessment of coronary artery disease, Fractional Flow Reserve (FFR) is widely used to determine whether a stenotic lesion produces functionally significant flow limitation. FFR is an invasive, wire-based physiological index measured during coronary angiography to quantify the pressure drop across a stenotic lesion. It is defined as the ratio of mean distal coronary pressure to mean aortic pressure during maximal hyperaemia, which is induced pharmacologically to assess the practical importance of the narrowing. In clinical practice, an FFR value of ≤ 0.80 is widely used as the threshold indicating functionally significant ischaemia and therefore provides an important guide for revascularisation decisions [1].

Although FFR is widely regarded as the clinical reference standard, its invasive nature introduces both procedural risk and cost. This has encouraged the development of non-invasive alternatives, among which Computational Fluid Dynamics (CFD) has emerged as a particularly important approach [2]. CFD frameworks, such as those used in FFR-CT systems, apply numerical techniques to resolve the governing equations of fluid flow within patient-specific anatomical models reconstructed from medical imaging. Through virtual simulation of coronary haemodynamics, these approaches make it possible to estimate pressure loss and derive FFR non-invasively [2].

However, the reliability of CFD-derived FFR depends strongly on modelling assumptions and data preparation [2]. Factors such as boundary conditions, assumptions about blood rheology, and the geometric representation of the coronary anatomy can all influence the resulting estimates. In particular, the way stenosis severity is defined is highly important. A severity definition based on vessel diameter reduction is not equivalent to one based on cross-sectional area reduction, and these two formulations can produce systematically different physiological outcomes. As demonstrated in this project, even subtle inconsistencies in the definition of anatomical parameters can introduce significant bias into simulation-derived FFR values and undermine their physiological validity [2].

2.2 Generative Adversarial Networks (GANs)

Generative Adversarial Networks (GANs), introduced by Goodfellow et al. [3], are among the most significant classes of deep generative models. A GAN consists of two neural networks trained simultaneously in an adversarial setting: a Generator (G) and a Discriminator (D) [3].

The generator is designed to learn the distribution of a target dataset. It takes a random noise vector as input and transforms it into a synthetic sample that aims to resemble the real data. The discriminator, by contrast, acts as a binary classifier that attempts to differentiate real samples from synthetic samples generated by the generator [3].

During training, the generator becomes increasingly more adept at producing realistic outputs that can deceive the discriminator, while the discriminator improves at identifying generated forgeries. This competitive process encourages both networks to improve over time. Ideally, the generator learns to approximate the real data distribution, and the discriminator becomes a powerful feature extractor capable of recognising the characteristics of authentic data in the relevant domain [4].

Despite their success, GANs are well known to be challenging to train. Common difficulties include instability, mode collapse, and sensitivity to architectural and optimisation choices [3], [5]. These issues are particularly relevant in structured or tabular data settings, where the relationships between variables may be subtle and where training can easily converge to trivial or poorly calibrated solutions if the adversarial balance is not carefully controlled.

2.3 Conditional GANs (cGANs)

A limitation of the original GAN framework is that it does not provide direct control over the type of data being generated [3]. Conditional Generative Adversarial Networks (cGANs) address this by supplying both the generator and the discriminator with additional conditioning information, usually denoted by y [4]. In this setup, the generator learns to produce samples conditioned on y, while the discriminator learns to determine whether a sample is realistic given the same condition [4].

This conditional structure allows the model to learn not just a general data distribution, but a conditional distribution relative to specific context variables [4]. In the context of this project, the conditioning vector consists of anatomical and lesion-related descriptors such as artery diameter, stenosis severity, lesion length, and derived haemodynamic features. The cGAN therefore learns the distribution of plausible FFR values given a particular anatomical context.

This is especially important here because an FFR value cannot be assessed meaningfully in isolation [1], [2]. Its plausibility depends on the geometry of the artery and the severity and structure of the stenosis with which it is associated [2]. A conditional formulation is therefore more appropriate than a standard GAN, because it enables the discriminator to assess whether an FFR value is physiologically consistent with the anatomical conditions under which it occurs.

2.4 GANs for Anomaly and Out-of-Distribution Detection

Although GANs are primarily known for generative modelling, their architecture is also well suited to anomaly detection and out-of-distribution (OOD) detection [8]. The central idea is that a discriminator trained on a specific reference distribution can learn the patterns that define normal or in-distribution data. At test time, this learned representation can be used to identify samples that deviate from that distribution.

In such settings, the discriminator is expected to assign higher scores to samples that conform to the distribution on which it was trained and lower scores to samples that fall outside it. This makes the discriminator a potentially useful scoring function for OOD detection. Several GAN-based anomaly detection methods have been proposed in the literature, including AnoGAN and GANomaly, both of which train adversarial models on normal data and then use either reconstruction behaviour or discriminator-derived signals to identify anomalous samples at test time [6], [7]. However, existing methods such as AnoGAN and GANomaly were developed for image-based anomaly detection and do not natively handle the conditional tabular structure of FFR-anatomy pairs, motivating the conditional GAN formulation adopted in this work.

The present project adapts this principle to the physiological assessment of FFR data. Here, the "normal" or in-distribution data correspond to the trusted reference dataset, while the "anomalies" correspond to physiologically inconsistent samples from the biased dataset. Rather than using the GAN purely for generation, the discriminator is repurposed as a physiological plausibility scorer. A high score indicates that the joint pattern of anatomy and FFR resembles the trusted physiological reference manifold, while a low score suggests that the sample is out-of-distribution and therefore likely to reflect simulation bias.

This framing places the project at the intersection of conditional generative modelling and OOD detection. The contribution is not the invention of GAN-based anomaly detection itself, but its application as a physiology-aware quality control framework for distinguishing a trusted reference distribution from a biased simulation distribution.

3. Methodology

This section presents the methodology used to develop and evaluate the proposed conditional Generative Adversarial Network (cGAN) for physiological plausibility scoring of FFR values. The experimental design was structured to ensure that the model learned only from physiologically consistent reference data, while the biased dataset was reserved entirely for evaluation as an out-of-distribution set. The complete workflow consisted of dataset merging and cleaning, feature engineering, normalization and splitting, cGAN construction, unsupervised warm-up, and adversarial optimisation. After training, the discriminator was used as the final scoring function, assigning each FFR-anatomy pair a plausibility score in the interval [0,1]. These scores were then analysed using summary statistics, threshold-based out-of-distribution detection, and discrimination metrics. All experiments were implemented in Python using TensorFlow/Keras, with preprocessing and analysis performed using NumPy, pandas, scikit-learn, and Matplotlib in Google Colab.

3.1 Datasets and Preprocessing

This project used two datasets generated under different definitions of stenosis severity. The first, referred to as the reference dataset, defined stenosis severity using diameter reduction and was treated as the physiologically consistent ground truth. The second, referred to as the biased dataset, defined stenosis severity using cross-sectional area reduction and was considered potentially physiologically inconsistent. The reference dataset was used exclusively for model training, while the biased dataset was reserved entirely for post-training evaluation. This ensured that the model learned physiological consistency only from the trusted reference distribution and that evaluation on the biased dataset represented genuine out-of-distribution testing.

The data were provided in four CSV files: two FFR summary files and two stenosis design files. The FFR files contained case identifiers and FFR values, while the design files contained anatomical descriptors including artery length, vessel

diameter, stenosis position, stenosis length, stenosis severity percentage, lesion boundaries, and eccentricity. The FFR files were merged with the corresponding design files using a shared S-number key extracted from the case identifiers via an inner join. This merge step ensured that each final case contained both the FFR value and the corresponding anatomical and lesion descriptors required for conditional plausibility assessment.

After merging, numerical columns were converted to numeric format and invalid entries were coerced to missing values. FFR values outside the physiologically meaningful interval [0, 1] were removed, as retaining them in earlier experiments distorted the learned distribution and produced unstable discriminator behaviour. After filtering, the final dataset sizes were 1,066 reference samples and 1,200 biased samples.

3.2 Feature Engineering

To capture both lesion geometry and the relationship between anatomy and haemodynamic behaviour, a set of engineered features was constructed from the merged data. The base anatomical variables included artery diameter, stenosis severity percentage, stenosis severity fraction, stenosis diameter (computed as the product of vessel diameter and severity fraction), lesion length (computed as the difference between lesion end and start positions), and an eccentricity indicator.

Three physiologically motivated derived features were also included: an expected FFR, defined as a severity-based heuristic approximation; an FFR residual, defined as the difference between the observed and expected FFR; and an FFR-by-severity interaction term. These features provided the discriminator with richer contextual information about whether an observed FFR value was consistent with the associated lesion characteristics. The final conditioning vector passed to the model consisted of nine variables: diameter_mm, stenosis_severity_pct, severity_frac, stenosis_diameter_mm, lesion_length_mm, is_eccentric, expected_ffr, ffr_residual, and ffr_x_severity.

3.3 Data Splitting and Normalization

Only the reference dataset was split into training (70%), validation (15%), and test (15%) sets, yielding 746 training, 160 validation, and 160 test samples. The biased dataset was not used during training, hyperparameter tuning, or model selection, preserving the integrity of the OOD detection setting.

Feature scaling was applied using Min-Max normalisation, fitted exclusively on the reference training set. The same fitted scaler was then applied to the biased dataset to ensure no distributional information from the biased data leaked into the training process. All values were subsequently clipped to [0, 1] to align with the sigmoid-bounded outputs of the model.

3.4 Model Architecture

The framework was a conditional Generative Adversarial Network (cGAN) composed of two fully connected neural networks.

The generator received a concatenation of a 16-dimensional random noise vector and the nine-dimensional conditioning vector. It passed these through dense layers with LeakyReLU activations and batch normalisation [9], with a final sigmoid output layer constraining the generated FFR value to [0, 1]. The generator's role was not to produce clinically deployable FFR values, but to serve as an adversarial mechanism that improved the discriminator's ability to model the reference physiological distribution.

The discriminator received a concatenation of one FFR value and the nine-dimensional conditioning vector. It passed these through several dense layers with LeakyReLU activations and dropout regularisation [10], with a single sigmoid output neuron producing a plausibility score in [0, 1]. Higher scores indicated that the FFR-anatomy pair was consistent with the learned reference distribution; lower scores indicated likely physiological inconsistency.

3.5 Two-Phase Training Strategy

A two-phase training strategy was adopted to stabilise the discriminator and establish a robust physiological boundary before adversarial training began.

Phase 1 - Unsupervised Warm-up. The discriminator was pre-trained for 80 epochs on the reference training data alone using an unsupervised warm-up strategy. Negatives were constructed internally by decoupling FFR values from their anatomical context within each mini-batch. Two types of mismatched pairs were created: FFR values shuffled against fixed conditions, and conditions shuffled against fixed FFR values. Both retained individually realistic values but broke the physiological relationship between them. The training signal derived entirely from the internal coherence structure of the reference data, without external labels or exposure to the biased dataset. The loss combined binary cross-entropy with a mean squared error term that pulled real scores toward 1.0 for stability. In this sense, the warm-up stage was effectively

self-supervised, since the model learned conditional coherence from internally constructed matched and mismatched pairs rather than from externally annotated anomaly labels. After warm-up, the discriminator produced a mean score of 0.826 on matched reference pairs compared with 0.107 on mismatched pairs, providing a well-calibrated starting point for adversarial training.

Phase 2 - Adversarial Training. Full adversarial training was carried out for 400 epochs with a batch size of 32. The discriminator was trained on real reference samples, generator-produced conditional samples, and shuffled hard negatives, and was updated twice per generator update to reflect the project's dependence on a strong discriminator. Binary cross-entropy loss was used with soft labels of 0.95 for real samples and 0.05 for fake samples [11]. A one-sided MSE penalty was additionally applied to the discriminator loss, penalising any reduction in real sample scores below the target of 0.95 with a weight of 0.4. This penalty was critical for preserving the discriminator's calibration with respect to the reference distribution throughout adversarial training. Small Gaussian input noise with standard deviation 0.003 was added to discriminator inputs as an additional regularisation mechanism. Learning rates were set asymmetrically, with the discriminator at 8×10^{-5} and the generator at 2×10^{-4}, to prevent either component from dominating the optimisation prematurely.

3.6 Physiological Plausibility Scoring and Evaluation

Once training was complete, the generator was discarded. The trained discriminator was applied to all samples in the reference training, validation, and test sets, as well as to the full biased dataset, producing a plausibility score in [0, 1] for each FFR–anatomy pair. The discriminator output was interpreted as a physiological plausibility score, where values closer to 1 indicate stronger consistency with the learned reference distribution and values closer to 0 indicate likely out-of-distribution behaviour.

Evaluation proceeded in three stages. First, summary statistics (mean, standard deviation, minimum, maximum) were computed for each dataset split to assess distributional separation. Second, threshold-based OOD detection was performed using the 10th and 25th percentiles of the reference test score distribution as cutoff values; the proportion of biased samples falling below each threshold was then calculated. These thresholds were intended to simulate a practical decision rule for flagging suspicious outputs based on how far they fell below the lower tail of the trusted reference score distribution. Third, global discrimination performance was measured using the Area Under the Receiver Operating Characteristic Curve (AUC-ROC), providing a threshold-independent measure of the model's ability to separate reference from biased samples. In addition, robustness was assessed across multiple random seeds to determine whether the learned plausibility boundary and detection performance remained stable under different stochastic initialisations.

4 Results

This section presents the performance of the proposed conditional Generative Adversarial Network (cGAN) as a physiological plausibility scorer for FFR data. The main objective of the evaluation was to determine whether a discriminator trained only on the physiologically consistent reference dataset could assign high scores to in-distribution samples and low scores to samples from the biased dataset. The results are reported using summary statistics of the plausibility scores, threshold-based out-of-distribution (OOD) detection, and a global discrimination metric in the form of the Area Under the Receiver Operating Characteristic Curve (AUC-ROC).

4.1 Overall Performance Metrics

The final trained model produced the following plausibility score statistics:

- Reference training set: mean = 0.8230, standard deviation = 0.0547
- Reference validation set: mean = 0.8237, standard deviation = 0.0537
- Reference test set: mean = 0.8225, standard deviation = 0.0545
- Biased dataset: mean = 0.0944, standard deviation = 0.1467

The score distributions were therefore highly separated. The reference samples consistently received scores above 0.8 on average, whereas the biased samples received scores close to zero on average. This large gap suggests that the discriminator learned a strong boundary between physiologically plausible and implausible samples.

The stability of the mean scores across the reference training, validation, and test sets is also important. It indicates that the model was not merely memorising the training data, but was instead capturing a more generalisable physiological structure that extended consistently to unseen reference cases.

In addition to the score summaries, the final model achieved an AUC-ROC of 0.9988, demonstrating near-perfect discrimination between the reference and biased datasets. This confirms that the plausibility score was not only different on average between the two groups, but also highly effective as a ranking signal for distinguishing in-distribution from

out-of-distribution samples. Beyond AUC-ROC, the model achieved an AUC-PR of 0.9924, a KS statistic of 0.9696 ($p = 2.01 \times 10^{-173}$), an accuracy of 98.8%, sensitivity of 90.0%, specificity of 100%, precision of 100%, and an F1 score of 0.9474 at the 10th percentile threshold. Full evaluation outputs including the ROC curve and precision-recall curve are provided in Appendix C.

4.2 Plausibility Score Analysis

A closer inspection of the score distributions provides further insight into the behaviour of the model. The reference test samples formed a compact high-score distribution centred around approximately 0.823, whereas the biased samples were concentrated in a much lower-score region with a mean of 0.094.

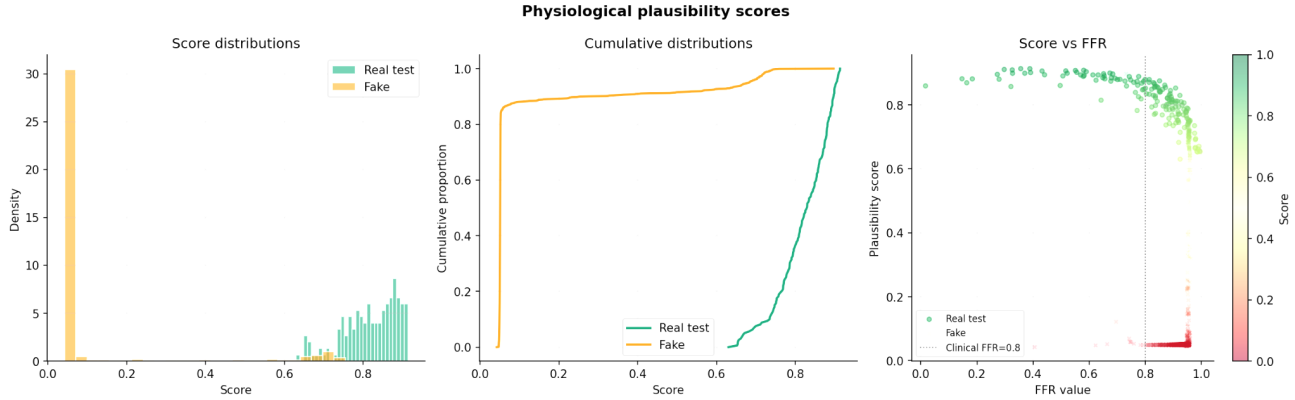


Figure 1: Physiological Plausibility Score

Figure 1 shows a clear separation between the plausibility scores assigned to the reference test set and those assigned to the biased dataset. In the left-hand panel, the reference samples are concentrated in a high-score region, while the biased samples are clustered much closer to zero. This confirms that the discriminator learned a strong distinction between physiologically consistent and inconsistent FFR-anatomy pairs.

The cumulative distributions in the middle panel reinforce this result by showing that the biased samples accumulate rapidly at very low scores, whereas the reference samples remain concentrated at substantially higher values. This indicates that the vast majority of biased cases fall outside the main reference score range. The right-hand panel further shows that the discriminator score remains high for most reference samples across the observed FFR range, while biased samples receive systematically lower scores, particularly in the high-FFR region where the biased dataset is most concentrated. Together, these patterns support the interpretation that the trained cGAN discriminator functions effectively as a physiological plausibility scorer and an out-of-distribution detector.

The biased samples were not simply assigned lower scores because they came from a different file or data source. Rather, they were scored according to how well their FFR values aligned with the anatomical conditions learned from the trusted reference set. The low mean score assigned to the biased dataset suggests that the model detected a systematic mismatch between the geometry-related inputs and the associated FFR values in these samples.

The reference test distribution also exhibited a moderate spread, with scores ranging from 0.628 to 0.884. This reflects the fact that not all physiologically valid reference cases are equally typical under the learned distribution. Some cases lie closer to the centre of the reference manifold, while others are more borderline but still remain substantially more plausible than the vast majority of biased samples.

By contrast, the biased dataset showed a lower mean and a wider relative spread, with scores ranging from 0.044 to 0.724. Although a small number of biased samples achieved intermediate plausibility values, the overwhelming majority remained far below the main reference distribution. This is consistent with the idea that the biased data contain FFR-anatomy relationships that deviate from the physiological patterns captured from the reference set. Additional score distribution plots including violin plots and box plots are provided in Appendix B. Feature-level analysis of plausibility scores across conditioning variables is provided in Appendix D.

4.3 Out-of-Distribution (OOD) Detection Performance

To evaluate the model as an operational OOD detector, thresholds were derived from the reference test score distribution. Two thresholds were selected: the 10th percentile and the 25th percentile of the reference test scores.

At the 10th percentile threshold:

- threshold value = 0.7568
- proportion of reference test samples below threshold = 10.0%
- proportion of biased samples below threshold = 100%

At the 25th percentile threshold:
- threshold value = 0.8057
- proportion of reference test samples below threshold = 25.0%
- proportion of biased samples below threshold = 100%

These results show that all biased samples were identified as out-of-distribution relative to the physiological reference manifold. At the 10th percentile threshold, 100.0% of biased samples fell below the cutoff. Under the 25th percentile threshold, all 1,200 biased samples were likewise rejected. This complete separation at both thresholds confirms that the learned physiological boundary cleanly encompasses the entire biased dataset, with zero false acceptances across either threshold.

This threshold-based analysis is particularly important because it demonstrates that the discriminator score is not only useful as a continuous ranking measure, but can also be used practically as a detection mechanism. In other words, the model can support a decision rule in which low-scoring samples are flagged for further review or rejection as physiologically inconsistent.

4.4 Training Behaviour

The training curves showed that the two-phase training strategy led to a stable and well-separated discriminator. During the warm-up stage, the discriminator learned to distinguish matched reference samples from internally shuffled mismatched pairs, producing a mean score of 0.826 on real matched data compared with 0.107 on shuffled negatives. This provided a strong and methodologically clean starting point for adversarial training.

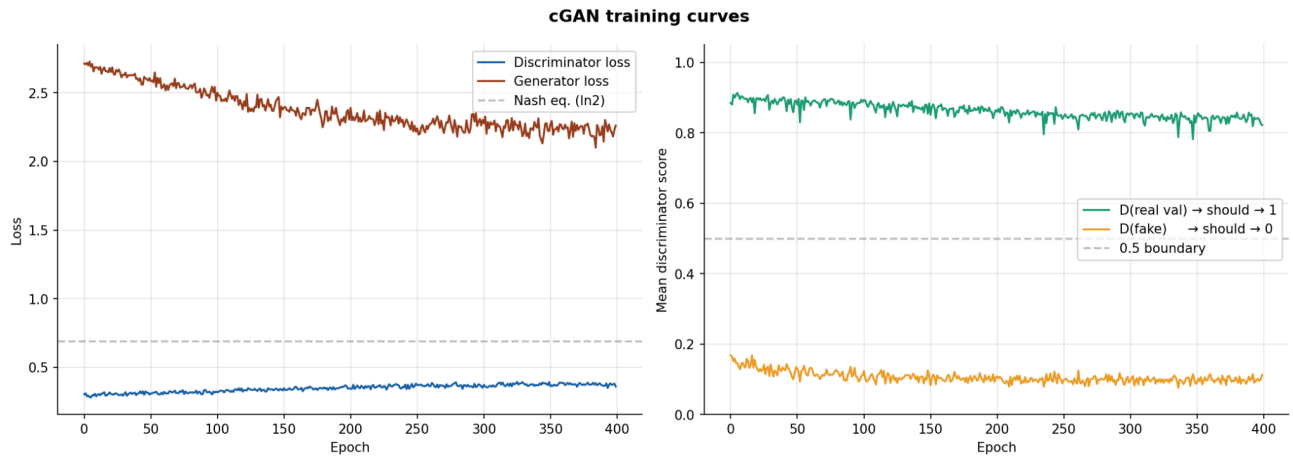


Figure 2: cGAN Training Curve

As shown in Figure 2, the discriminator loss remains relatively low and stable throughout training, while the generator loss stays higher, indicating that the discriminator maintained a strong ability to distinguish physiologically consistent samples from generated alternatives. The right-hand panel further shows that the discriminator assigned consistently high scores to the reference validation set and low scores to the biased dataset across epochs. This persistent separation suggests that the training process remained stable and that the model learned a meaningful physiological plausibility boundary rather than collapsing to a trivial solution.

During the adversarial phase, the discriminator continued to maintain high scores on the reference validation data and low scores on the biased dataset. By the end of training, the validation mean remained above 0.8 while the mean biased score remained close to 0.1.

Taken together, these findings show that the final cGAN configuration succeeded in learning a robust conditional plausibility function from the reference dataset and applying it effectively to out-of-distribution detection on the biased data. Extended training stability plots are provided in Appendix A.

4.5 Seed Stability Analysis

To further assess this conclusion, the final model was repeated across five independent random seeds (42, 7, 123, 99, and 2024). Performance remained consistently strong across all runs. The reference test set achieved a mean plausibility score of 0.8201 ± 0.0271, while the biased dataset achieved a mean score of 0.0944 ± 0.0109, corresponding to an average separation gap of 0.7257 ± 0.0203. Discrimination performance was also highly stable, with an average AUC-ROC of 0.9968 ± 0.0017 and an average AUC-PR of 0.9569 ± 0.0090. Using the 10th-percentile threshold derived from the reference test distribution, the model detected 99.48% ± 0.57% of biased samples as out-of-distribution. The lowest-performing run was seed 123, which produced a reference mean score of 0.769 due to a transient suppression of real sample scores around epoch 200, where D(val_real) dipped to 0.757 before partially recovering. This behaviour is consistent with a known instability in GAN training where adversarial dynamics temporarily erode discriminator confidence on real data. Importantly, the AUC-ROC for this run was 0.9975 - among the highest across all seeds - confirming that distributional separability between reference and biased samples was preserved throughout, even when absolute score calibration fluctuated. These results indicate that the proposed cGAN-based plausibility scorer is not dependent on a single favourable random seed and remains highly reproducible across repeated runs. This stability is attributable in part to the architectural choices described in Section 3.5, specifically the real score penalty, the asymmetric learning rates, and the two discriminator updates per generator step, which collectively preserved the discriminator's calibration even under unfavourable initialisations such as seed 123. Extended training stability analysis is provided in Appendix A.

5 Discussion

This section discusses the significance of the experimental findings and relates them back to the objectives of the project. The section also considers the limitations of the current framework and identifies possible directions for further development.

5.1 Interpretation of Results

The results indicate that the proposed conditional Generative Adversarial Network (cGAN) learned a meaningful notion of physiological plausibility from the reference data.

The mean plausibility score of 0.8225 on the reference test set, compared with 0.0944 on the biased dataset, suggests that the biased samples lie well outside the physiological manifold learned from the trusted data. In practical terms, the model learned what a physiologically coherent FFR-anatomy relationship looks like and assigned low scores when that relationship was violated.

The threshold-based results reinforce this interpretation. Since 100% of biased samples fell below both the 10th- and 25th-percentile thresholds derived from the reference test distribution, the discriminator score functions not only as a descriptive plausibility measure but also as an effective operational signal for identifying suspect samples. The very high AUC-ROC of 0.9988 further supports this conclusion by showing that the learned score provides near-perfect ranking separation between trusted and biased cases. The strength of these results is attributable to several key methodological choices, discussed in the following section.

Despite the strong results, several caveats apply. The sensitivity of 90.0% means that 10% of reference samples are rejected as implausible at the 10th percentile threshold; these borderline cases likely represent physiologically valid but atypical FFR-anatomy combinations that fall in the lower tail of the reference distribution. Additionally, the 100% biased detection rate should be interpreted cautiously: the bias in this dataset is systematic and pronounced, and the model may perform less well on more subtle forms of simulation inconsistency. These considerations motivate the future work directions outlined in Section 5.3.

5.2 Justification of Key Design Decisions

Several methodological choices proved critical to the success of the final model.

Use of the reference dataset only for training

The decision to train exclusively on the reference dataset was central to the validity of the approach. Because the project objective was to learn the distribution of physiologically consistent FFR values, including the biased dataset during training would have contaminated the model’s definition of plausibility. By reserving the biased dataset exclusively for evaluation, the learned score remained grounded in the trusted reference manifold. This also ensured that the final results represented genuine generalisation rather than memorisation of biased patterns.

Filtering invalid FFR values

A key turning point in the project was the removal of invalid reference FFR values, particularly negative values and other observations outside the physiologically meaningful range of [0,1]. Since FFR is defined as a pressure ratio, such values are non-physical and likely reflect data or simulation artefacts rather than valid haemodynamic behaviour. When these corrupted values were retained, the discriminator learned an unstable and distorted reference distribution, leading to nearly constant outputs and, in some cases, reversed separation between reference and biased datasets. Removing these invalid observations substantially improved the model's ability to learn a coherent physiological manifold. This result emphasises the critical role of data quality and preprocessing in adversarial learning for biomedical plausibility assessment.

Enriched feature engineering
The addition of derived physiological features such as expected FFR, FFR residual, and the FFR–severity interaction term also contributed substantially to final performance. These variables provided the discriminator with richer information about whether an observed FFR was compatible with the lesion characteristics. Rather than depending only on raw geometry, the model benefited from a more informative feature space that reflected both anatomy and expected haemodynamic behaviour.

Unsupervised warm-up
The use of unsupervised warm-up before full adversarial optimisation was another key design choice. This stage allowed the discriminator to first learn the difference between correctly matched reference samples and shuffled mismatched samples, without the added instability of generator competition. The resulting initial separation provided a much stronger starting point for subsequent adversarial training. In practice, this helped avoid the failure modes seen in earlier experiments, where the discriminator either collapsed to near-constant scores or learned a notion of "realness" that did not align with physiological consistency. Crucially, this warm-up used no external labels, negatives were generated entirely by internal shuffling of the reference dataset, making the approach self-supervised rather than supervised.

Hard negative sampling
The inclusion of shuffled FFR-anatomy pairs as hard negatives was especially valuable. These samples preserved realistic individual values while breaking the physiological relationship between them. As a result, the discriminator was forced to learn conditional compatibility rather than simply learning whether an isolated FFR number looked typical. This made the training objective much closer to the actual research problem, namely whether a given FFR value is plausible for a given anatomy.

Real score penalty
A one-sided MSE penalty was applied during adversarial training to prevent the discriminator from losing confidence about real reference samples over time. This penalty fired whenever real sample scores fell below the target of 0.95, directly counteracting the tendency of adversarial dynamics to erode the discriminator's calibration with respect to in-distribution data. Without this mechanism, earlier experiments showed that real sample scores gradually declined during adversarial training, narrowing the separation gap and reducing detection performance. The penalty was critical for maintaining the stable, high reference scores observed in the final model.

Several earlier configurations were explored before arriving at the final design. Initial experiments retained negative FFR values in the reference dataset, which produced a distorted learned distribution and near-constant discriminator outputs. A supervised pretraining stage using binary labels was also tested but was found to violate the methodological constraint of training without biased data exposure. The unsupervised warm-up strategy was adopted as a principled alternative that preserved the self-supervised nature of the approach.

5.3 Limitations and Future Work
Despite the strength of the final results, the study has several limitations.

Dependence on simulated data
First, the entire framework was developed and evaluated using datasets rather than invasive clinical measurements. Although the reference dataset was treated as physiologically consistent, it remains the product of simulation assumptions rather than direct patient-grounded haemodynamic truth. This means that the learned plausibility manifold reflects the quality and assumptions of the simulation pipeline itself.

Limited scope of anatomical representation
Second, the model was trained on a limited set of anatomical and engineered tabular descriptors. While these variables capture important lesion and vessel characteristics, they do not represent the full richness of coronary geometry or

haemodynamic flow fields. More detailed CFD outputs, such as pressure or velocity field summaries, might provide additional discriminatory information and could improve future models.

Heuristic engineered features
Third, some of the engineered features, particularly expected FFR, were based on simplified heuristics rather than exact physiological formulas. These features proved useful in practice, but they are not substitutes for a full mechanistic haemodynamic model. A future extension could replace or augment them with more rigorously derived descriptors grounded in cardiovascular fluid mechanics.

Dataset-specific evaluation
Fourth, the evaluation was performed on one specific biased dataset constructed under a known alternative definition of stenosis severity. While the results show excellent discrimination in this setting, it is not yet known how the model would perform on other forms of simulation bias or on external datasets generated by different pipelines. Further validation on independent datasets, and ideally on clinically derived data, would be necessary to establish broader generalisability.

Calibration and interpretability
Finally, although the discriminator score behaved as a useful plausibility measure, it should not be interpreted as a clinically calibrated probability. The score is best understood as a relative measure of consistency with the learned reference distribution. Future work could investigate explicit score calibration, uncertainty quantification, or conformal-style thresholding methods to improve interpretability and deployment readiness.

Future directions
Several directions for future work emerge from this study. One possibility is to compare the cGAN approach with simpler real-only OOD methods such as one-class classifiers, density-based models, or autoencoder-based anomaly detectors. Another is to incorporate richer geometry representations or graph-based vessel models. A further extension would be to test whether the same framework can detect more subtle forms of simulation bias, not only different stenosis definitions but also changes in boundary conditions, mesh assumptions, or haemodynamic parameters. Ultimately, applying the framework to clinically validated patient-specific data would be the most important step toward practical translation.

5.4 Overall Significance

From a broader perspective, the main contribution of this work is both conceptual and technical. Rather than using deep learning purely as a predictive tool, this project applies conditional adversarial learning as a quality-control mechanism for simulation-derived biomedical data. The model does not attempt to replace CFD or directly correct biased outputs; instead, it learns the structure of a trusted physiological distribution and identifies samples that deviate from it.

This is important because the growing use of simulation and AI in healthcare increases the need for systems that assess the trustworthiness of computational outputs, not just generate them. In this sense, the proposed cGAN-based plausibility scorer contributes to the wider goal of making computational medicine safer and more reliable.

6 Conclusion

This project presented a conditional Generative Adversarial Network (cGAN) framework for assessing the physiological plausibility of Fractional Flow Reserve (FFR) values and detecting biased samples as out-of-distribution. The motivation was the presence of two simulation datasets generated under different definitions of stenosis severity: a trusted reference dataset based on diameter reduction and a biased dataset based on cross-sectional area reduction. Because these definitions are not physiologically equivalent, the objective was not to correct biased values directly, but to learn the conditional distribution of physiologically consistent FFR values from the reference dataset and use this learned representation to identify inconsistent samples.

The final model achieved a mean reference test score of 0.8225 against a biased mean of 0.0944, with 100% of biased samples detected at the 10th percentile threshold and an AUC-ROC of 0.9988. Seed stability testing across five independent runs confirmed reproducibility, with a mean AUC-ROC of 0.9968 ± 0.0017.

Overall, this study demonstrates that learning physiological plausibility from trusted reference data is both feasible and highly effective for detecting simulation bias in FFR datasets. The framework requires no knowledge of the bias in advance and no exposure to biased data during training, making it applicable to quality-control scenarios where the nature of the inconsistency is unknown. Future work should extend validation to patient-specific anatomies and alternative forms of simulation bias to establish broader clinical applicability.

7 References

[1] P. A. L. Tonino et al., "Fractional Flow Reserve versus Angiography for Guiding Percutaneous Coronary Intervention," New England Journal of Medicine, vol. 360, no. 3, pp. 213–224, Jan. 2009.
[2] C. A. Taylor, T. A. Fonte, and J. K. Min, "Computational Fluid Dynamics Applied to Cardiac CT for Noninvasive Quantification of Fractional Flow Reserve: Scientific Basis," Journal of the American College of Cardiology, vol. 61, no. 22, pp. 2233–2241, Jun. 2013.
[3] I. Goodfellow et al., "Generative Adversarial Nets," in Advances in Neural Information Processing Systems 27, 2014, pp. 2672–2680.
[4] M. Mirza and S. Osindero, "Conditional Generative Adversarial Nets," arXiv preprint arXiv:1411.1784, 2014.
[5] M. Arjovsky, S. Chintala, and L. Bottou, "Wasserstein Generative Adversarial Networks," in Proceedings of the 34th International Conference on Machine Learning (ICML), 2017, pp. 214–223.
[6] T. Schlegl, P. Seeböck, S. M. Waldstein, U. Schmidt-Erfurth, and G. Langs, "Unsupervised Anomaly Detection with Generative Adversarial Networks to Guide Marker Discovery," in Information Processing in Medical Imaging (IPMI), 2017, pp. 146–157.
[7] S. Akcay, A. Atapour-Abarghouei, and T. P. Breckon, "GANomaly: Semi-Supervised Anomaly Detection via Adversarial Training," in Asian Conference on Computer Vision (ACCV), 2018, pp. 622–637.
[8] J. Yang et al., "Generalized Out-of-Distribution Detection: A Survey," arXiv preprint arXiv:2110.11334, 2021.
[9] S. Ioffe and C. Szegedy, "Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift," in Proceedings of the 32nd International Conference on Machine Learning (ICML), 2015, pp. 448–456.
[10] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A Simple Way to Prevent Neural Networks from Overfitting," Journal of Machine Learning Research, vol. 15, pp. 1929–1958, 2014.
[11] R. Müller, S. Kornblith, and G. Hinton, "When Does Label Smoothing Help?" in Advances in Neural Information Processing Systems 32, 2019, pp. 4694–4703.

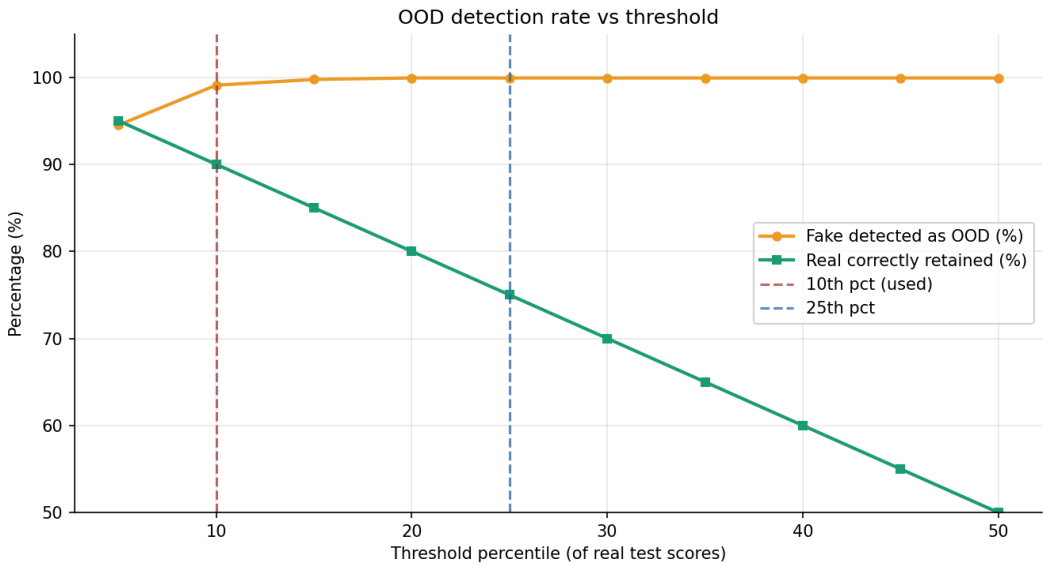
8 Appendix

This appendix provides supplementary figures that support the training analysis, score distribution analysis, evaluation metrics, and exploratory feature-level interpretation of the proposed cGAN-based plausibility framework.

Appendix A.

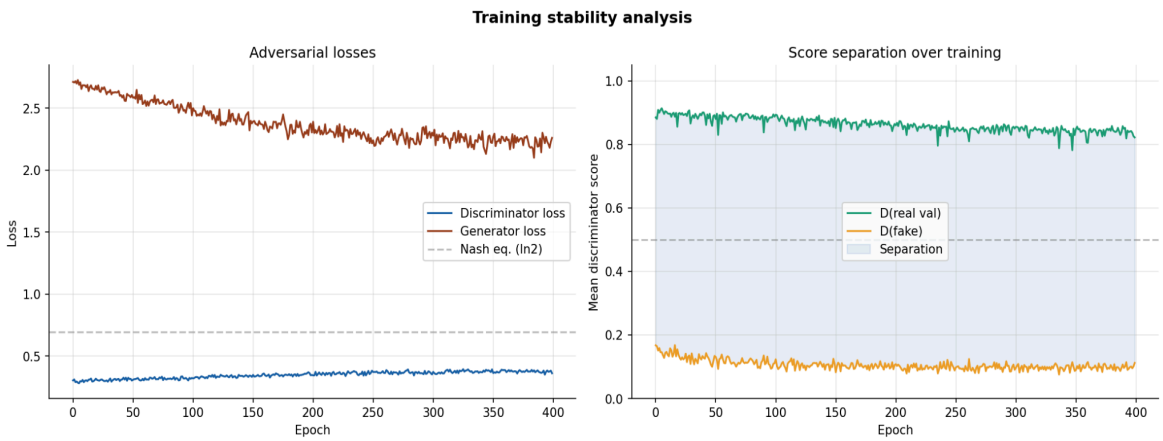
Additional Training and Stability Analysis

Figure A1. OOD detection rate as a function of threshold percentile



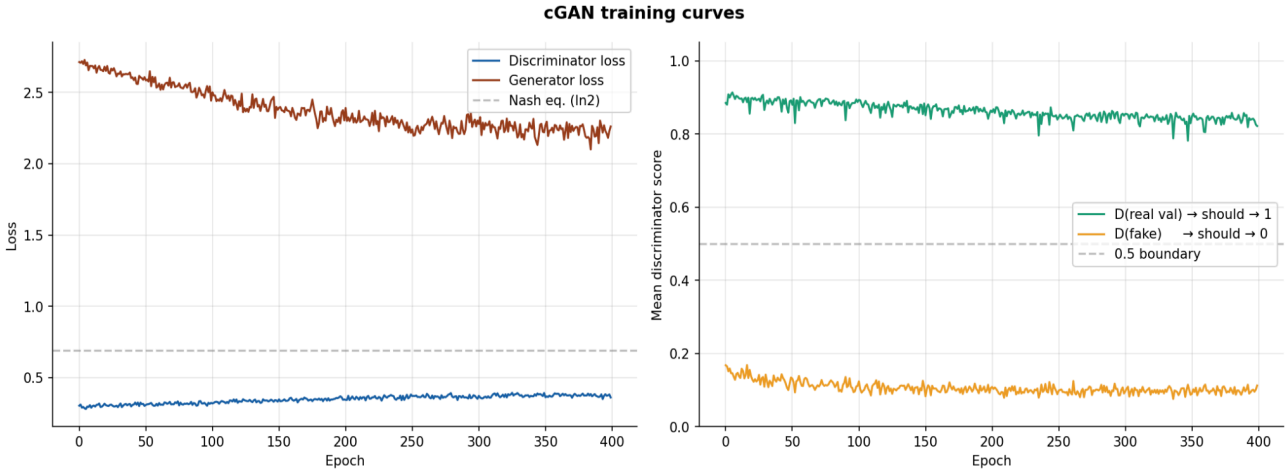
This figure shows how the proportion of biased samples detected as out-of-distribution changes as the decision threshold is varied across percentiles of the reference test score distribution. It also shows the proportion of reference samples correctly retained. The selected 10th- and 25th-percentile thresholds are marked for comparison.

Figure A2. Extended training stability analysis



This figure provides additional evidence of training stability. The left panel shows discriminator and generator loss across training epochs, while the right panel shows the mean discriminator scores assigned to the reference validation set and the biased dataset. The persistent separation indicates stable adversarial learning.

Figure A3. Alternative cGAN training curves

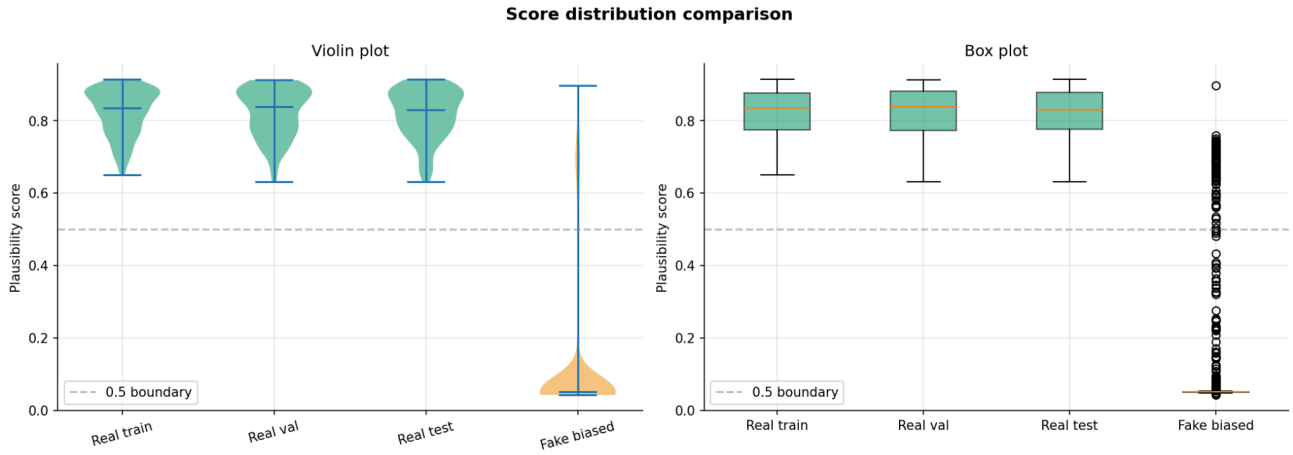


This figure presents an additional view of adversarial training behaviour, including the discriminator and generator losses and the evolution of discriminator scores on real validation and biased samples. It supports the conclusion that the final model learned a stable physiological plausibility boundary.

Appendix B.

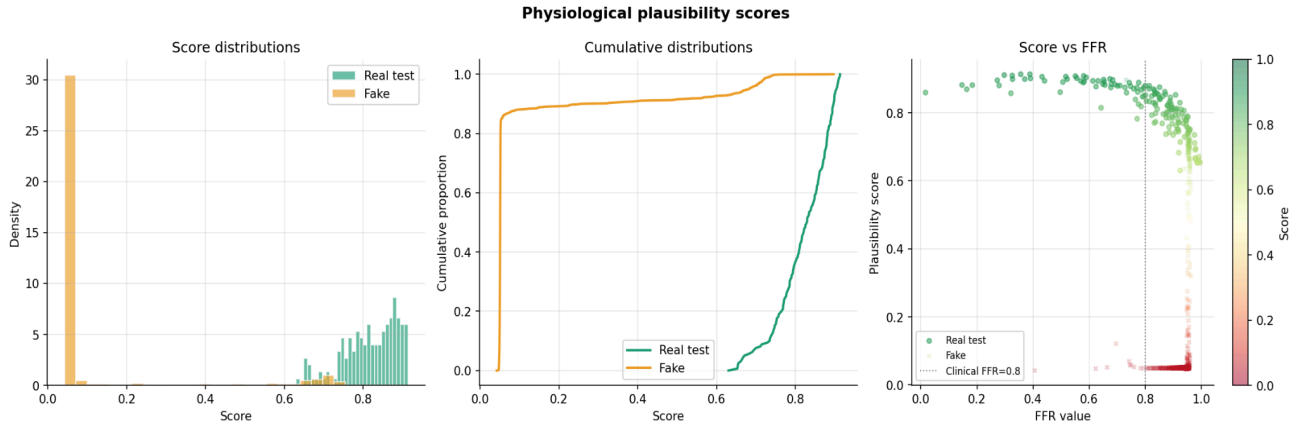
Additional Score Distribution Analysis

Figure B1. Violin and box plot comparison of plausibility scores



This figure compares the plausibility score distributions for the reference training, validation, test, and biased datasets using violin and box plots. It provides an alternative visual summary of the separation between physiologically consistent and biased samples.

Figure B2. Extended physiological plausibility score distributions

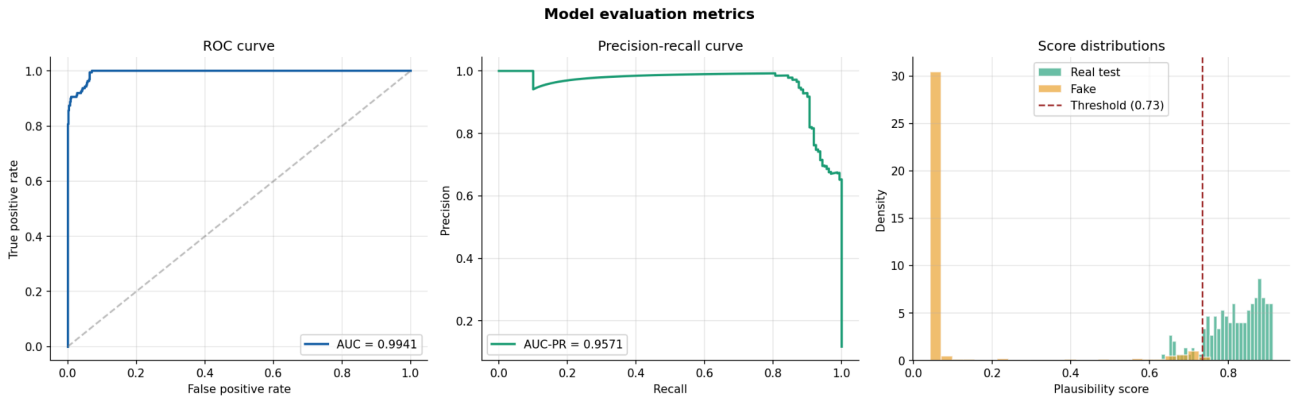


This figure shows additional views of the final plausibility scores, including the score histograms, cumulative score distributions, and the relationship between score and FFR value. It complements the main text by illustrating the consistency of the learned score behaviour.

Appendix C.

Additional Evaluation Metrics

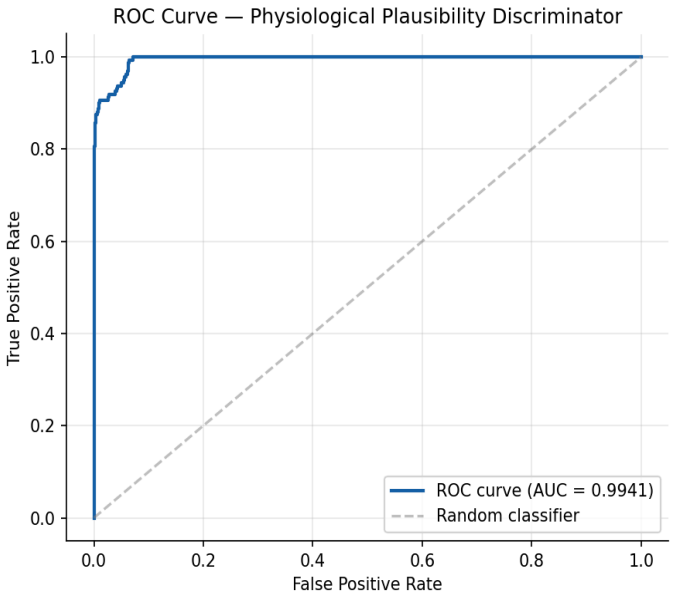
Figure C1. Extended model evaluation metrics



This figure summarises additional evaluation outputs, including the ROC curve, precision–recall curve, and score histogram with a threshold marker. It provides further evidence of strong discrimination between reference and biased samples.

Figure C2. ROC curve for the physiological plausibility discriminator

This figure shows the receiver operating characteristic curve for the final discriminator, with an AUC-ROC close to 1 indicating excellent separation between reference and biased samples.

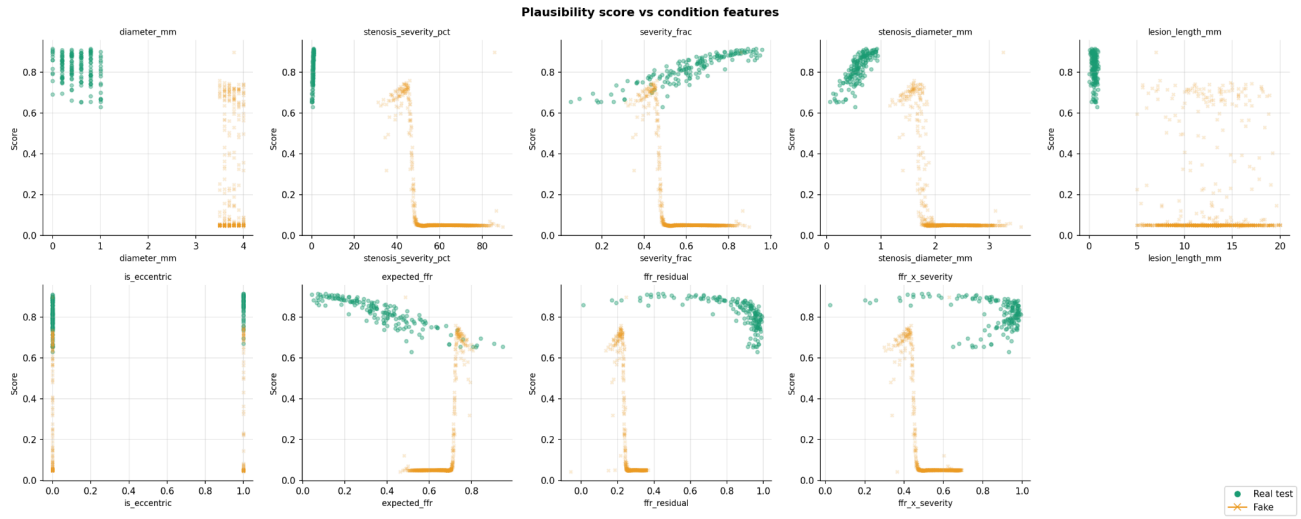


Appendix D.

Additional Feature and FFR Relationship Analysis

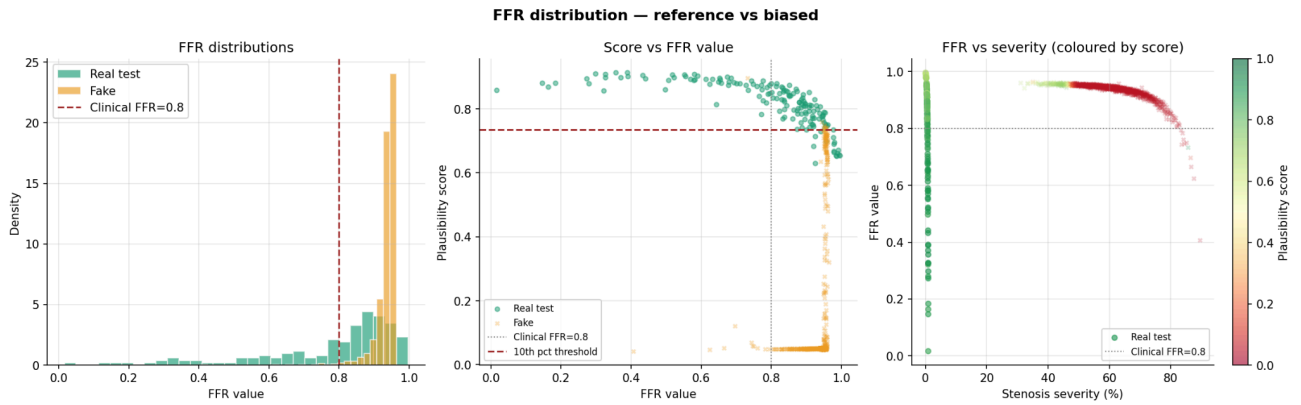
Figure D1. Plausibility score versus condition features

This figure plots plausibility scores against the main conditioning features used by the model. It illustrates how the



discriminator score varies across anatomical and engineered descriptors for reference and biased samples.

Figure D2. FFR distribution: reference versus biased



This figure compares the FFR distributions of the reference and biased datasets and relates them to plausibility score and stenosis severity. It provides additional context for understanding why the biased dataset is detected as out-of-distribution.